

Shenzhen SPV Field IoT Co., Ltd.

Attn.: Ms. Emily Huang (Sales)

Guangdong, China

via Alibaba Messenger

Munich, Germany — 26 April 2026

**Subject: Quotation Request — Custom-Printed Round Stickers Combining QR Code and Pre-Programmed NFC NTAG, in Multiple Sizes for Indoor & Outdoor Plant Tracking (Pilot 1,000 pcs, follow-up orders planned)**

Dear Ms. Huang, dear SPV Field team,

we are Planum, a German consumer-tech startup launching a smart plant-tracking system. Our customers buy a beautifully designed set of weatherproof QR + NFC stickers, attach them to the pots of their houseplants, garden plants and outdoor planters, and connect each plant to its individual digital care profile in our web app.

We selected your company because of your strong track record in printed NFC stickers (157,000+ units sold), your in-house QR + NFC programming capability and your existing experience with NFC plant/tree identification tags. We believe you are an ideal fit for our long-term sticker programme.

This enquiry concerns an initial pilot / sample production. Following a successful pilot we plan to scale significantly with follow-up orders (target volume 10,000 – 50,000 pcs per order).

## 1. Product Description — One SKU, Three Sizes

We require ONE single product specification, where every sticker combines a printed unique QR code AND a pre-programmed NFC tag in the same physical unit. There are NO separate QR-only or NFC-only versions. End-customers freely choose whether they connect their plant via smartphone scan (QR) or NFC tap.

To cover our full product range — houseplants, garden plants, outdoor planters and herbs — we want to offer the same sticker in three diameters within ONE end-customer pack:

- ø 25 mm — small pots, herbs, seedlings, propagation cups
- ø 30 mm — standard houseplants (default size)
- ø 40 mm — garden plants, outdoor planters, large pots

All three sizes share IDENTICAL material, finish, NFC chip, durability and certifications. Only the diameter differs. This is critical for us: one SKU, one production line, three diameters.

Important — the QR code and the NFC chip must be programmed with **TWO DIFFERENT URLs** per sticker. We will deliver a CSV with three columns (uuid, qr\_url, nfc\_url) per unit. Background: the QR code points to a public registration / login page, while the NFC chip points to a secure auto-login URL with a per-sticker token. This is essential for our user experience and is a hard product requirement.

## 2. Weather Resistance — Hard Requirement

Because our stickers will be applied not only to indoor pots but also to outdoor planters, raised garden beds and balcony pots, all stickers (regardless of size) **MUST** be fully weatherproof for at least 3 years of mixed indoor / outdoor use. Specifically:

- Permanent water resistance — repeated watering, rain, condensation, full submersion  $\geq 24$  h without delamination, ink running or QR-scan failure
- UV stability — Blue Wool Scale  $> 6$ , no visible colour fade after 12 months direct sunlight (south-facing window or garden)
- Temperature cycling —  $-20$  °C to  $+80$  °C without adhesive failure, brittleness or peeling
- Soil contact / dirt — wipeable, no permanent staining
- Adhesion on terracotta, glazed ceramic, plastic, glass and powder-coated metal

Material recommendation: we strongly prefer **PET** over PVC because of better long-term outdoor durability and lower environmental footprint. If you can offer a more weather-resistant alternative (e.g. heavier PET, synthetic paper), please propose.

### 3. Technical Specifications

| Parameter                        | Specification                                                                                          |
|----------------------------------|--------------------------------------------------------------------------------------------------------|
| Shape                            | Round, die-cut                                                                                         |
| Diameters (one SKU, three sizes) | $\varnothing 25$ mm / $\varnothing 30$ mm / $\varnothing 40$ mm                                        |
| Face material                    | PET, $\sim 80$ – $100$ $\mu$ m (PVC or synthetic paper as alternative — please advise)                 |
| Adhesive                         | Permanent acrylic, solvent-free, initial tack $> 8$ N/25 mm, suitable for porous surfaces (terracotta) |
| Temperature range                | $-20$ °C to $+80$ °C (operational)                                                                     |
| Printing                         | CMYK 4-colour, $\geq 600$ dpi (UV-digital or offset — please advise on best fit for our artwork)       |
| Finishing                        | Matte lamination + UV protective overlamine or varnish                                                 |
| Water resistance                 | $\geq 24$ h immersion without damage; permanent splash & rain resistance                               |
| Light fastness                   | Blue Wool Scale $> 6$                                                                                  |
| Durability target                | Minimum 3 years indoor + outdoor                                                                       |
| QR code                          | QR Model 2, Error Correction Level H (30 %), module size $\geq 0.5$ mm                                 |
| Variable QR data                 | Unique URL per sticker (qr_url column from our CSV)                                                    |
| NFC chip                         | NXP NTAG 213 (180 byte) as default — please also quote NTAG 215 (504 byte)                             |
| NFC frequency / protocol         | 13.56 MHz, ISO/IEC 14443A                                                                              |
| NFC programming                  | Pre-programmed unique URL per chip (nfc_url column from our CSV), locked against re-writing            |
| Certifications                   | REACH, RoHS, EU 10/2011, NFC chip CE / FCC                                                             |

**Material flexibility:** the parameters above are our research-based proposal. If your portfolio offers an equivalent or superior alternative (e.g. different vinyl thickness, alternative laminate, alternative NFC chip family), we are explicitly open to your counter-proposal — provided water resistance, UV stability and 3-year durability are met or exceeded.

## 4. Packaging — We Need Your Input

Each end-customer pack contains 20 stickers as a mixed assortment of the three sizes. Our preferred composition:

- 8 × ø 25 mm (small pots, herbs, propagation)
- 8 × ø 30 mm (standard houseplants)
- 4 × ø 40 mm (garden / outdoor)

Total per pack: 20 stickers · Pilot order: 50 packs · Pilot total: 1,000 stickers (400 × 25 mm + 400 × 30 mm + 200 × 40 mm).

### 4.1 Custom Outer Packaging — Open Questions

This is the area where we most need your guidance. Please answer the following points in your reply:

1. Custom packaging design — can the end-customer pack be produced from artwork supplied by us (paper sleeve / folded carton / box with Planum branding)? Or do you only supply standard packaging?
2. Existing packaging templates — do you offer ready-to-customise packaging templates (sleeves, folded cards, small boxes) we could choose from? If yes, please send dieline references / mockups.
3. Accepted artwork file formats — which file formats do you accept for print-ready artwork? (We can deliver AI, PDF/X-1a, EPS, vector SVG. Please confirm preferred DPI, bleed and colour profile — e.g. Fogra39.)
4. Carrier sheet — can the 20 stickers be supplied on an A6 carrier / liner sheet pre-arranged by size, rather than on a roll? This is essential for our gift-style unboxing experience.
5. Instruction insert — can you also print and insert a 1-page instruction card (DIN A6, 120 gsm, double-sided, CMYK) into each pack, again from artwork we supply?
6. Sustainable materials — preference for FSC-certified paper for sleeves / carrier and plastic-free outer wrap. Is this available, and at what surcharge?
7. Outer shipping carton — 50 end-customer packs per shipping carton, lot number and production date printed on the outside. Can you handle this?

## 5. Quantities — Pilot Order

| Item                     | Quantity                                   |
|--------------------------|--------------------------------------------|
| Total stickers (one SKU) | 1,000 pcs                                  |
| End-customer packs       | 50 packs of 20 stickers                    |
| Size mix per pack        | 8 × ø25 mm + 8 × ø30 mm + 4 × ø40 mm       |
| Size mix total           | 400 × ø25 mm + 400 × ø30 mm + 200 × ø40 mm |

|                                     |                                                            |
|-------------------------------------|------------------------------------------------------------|
| <b>Status</b>                       | Pilot / sample run; follow-up orders 10k – 50k pcs planned |
| <b>Target landed cost / sticker</b> | ≤ €0.55 incl. packaging, programming, freight, duties      |

We are aware that 1,000 pcs may be at or below your standard MOQ. Your Alibaba listing indicates MOQ 1,000 — please confirm. In return for accepting this pilot quantity, we commit to placing follow-up orders with you provided quality and delivery meet our expectations.

## 6. Mandatory Pre-Production Test Pack & Timeline

Before we release the main initial order, we require a fully assembled test pack (one complete end-customer set: 20 stickers in the size mix above on the carrier sheet, branded sleeve and instruction insert in the original packaging) for internal QA. We will test print quality, colour fidelity, water resistance, UV stability, adhesion on terracotta and the unboxing experience. Only after written approval will we release the main production. Our Amazon launch follows immediately afterwards.

- Test pack (1–2 fully packaged sets), via DHL Express: within 14 days of order confirmation
- Planum internal quality testing: ~7 days
- Written approval → start of main production
- Main batch lead time: max. 30 days from approval
- Main shipment: please quote both CIF Hamburg and DDP Germany

## 7. Shipping — Current Quote Required

The pilot shipment is small (~2 kg gross). Your Alibaba listing only shows ocean freight via Maersk (24–31 days). For a 2 kg pilot shipment, ocean freight is impractical. Please quote ALL of the following options and explicitly state current April 2026 rates and lead times:

- DHL Express / FedEx / UPS air courier — door-to-door (DDP Germany), incl. customs clearance and 19 % import VAT pre-paid
- Air freight forwarder — CIF Hamburg airport
- Sea-air mixed — if cost-effective at 2 kg
- Sea LCL via Maersk — CIF Hamburg port (only if substantial savings vs. air courier)
- Shipping insurance — included or separate?
- For follow-up orders at 10k / 25k / 50k pcs (~25 / 60 / 130 kg gross): please also quote sea LCL and air freight

## 8. Payment Terms

- Preferred: Alibaba Trade Assurance (full pilot order)
- Alternative: 30 % deposit on order, 70 % against B/L copy or before shipment
- Warranty: replacement or credit note for defect rates > 2 %

## 9. Itemised Quotation Request

We kindly ask you for a binding quotation including the following line items, separated and clearly labelled:

- 
- Unit price per sticker — for ø25 mm, ø30 mm and ø40 mm separately, all with QR + NFC NTAG 213 pre-programmed, PET face material
  - Per-unit surcharge for upgrading the NFC chip to NTAG 215 (504 byte)
  - Per-unit surcharge for two-URL programming (QR + NFC each with a different URL from our CSV)
  - One-off setup costs — tooling, die-cuts (3 sizes), printing plates, NFC programming setup
  - Custom packaging — sleeve / carrier sheet / instruction insert, per pack and total
  - Outer shipping carton — branded or plain, per carton
  - Shipping — all options from section 7 above
  - Lead times — test pack and main production
  - Official MOQ and tiered pricing for follow-up orders: 5k / 10k / 25k / 50k pcs
  - Reference projects — 2–3 examples of comparable QR + NFC sticker projects you have produced
  - Sample of a current production (any QR + NFC sticker from your range) included with the test pack, for reference

## 10. Attachments & Next Steps

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Attached to this enquiry is the full Planum sticker concept document (Planum\_Konzept\_Sticker\_v2.pdf) with brand guidelines, three artwork variants and detailed visuals. Print-ready vector files (AI / PDF/X-1a) will follow after your initial response and confirmation of accepted file formats.

We would be very pleased to receive your reply within the next 7 working days. Please feel free to contact us at any time via e-mail or Alibaba Messenger if anything is unclear.

Kind regards,

**Felix Bredl**

Founder · Planum

felixbredl@gmx.de

# PLANUM

## Smart QR + NFC Stickers — Product Concept v2

One SKU • Three Sizes (25 / 30 / 40 mm) • QR + NFC Combined • Weatherproof Indoor & Outdoor • Two-URL Programming for Auto-Login

This document describes the revised Planum sticker product for the manufacturer enquiry to Shenzhen SPV Field IoT Co., Ltd. It supersedes the v1 concept dated 25 April 2026.

Key changes vs. v1:

- ONE single product variant (SKU) — every sticker combines QR + NFC; no separate variants
- THREE diameters within ONE end-customer pack (25 / 30 / 40 mm) for indoor, herbs and outdoor use
- Hard weather-resistance requirement — outdoor / garden usage is now in scope from day one
- Two-URL programming — QR and NFC each carry a different URL per sticker (registration vs. auto-login)
- Material preference shifted from PP-Vinyl to PET (better outdoor durability, lower environmental footprint)

### 1 • Product Concept

Planum stickers are the physical bridge between a real plant and its digital care profile. Each sticker carries a unique printed QR code AND a unique pre-programmed NFC tag. The customer attaches the sticker to the plant pot — from that moment, the plant is part of their digital garden in our web app.

**Functional** — QR scans must be reliable for at least 3 years across watering, sunlight, soil contact, indoor heating and outdoor weather (rain, frost, summer heat, UV).

**Aesthetic** — On a beautiful pot the sticker should not feel like a label, but like a discreet brand mark.

**Recognisable** — Planum brand language: calm green tones, botanical motif, clean typography.

**Universal** — One product line covers houseplants, herbs, garden plants, balcony planters and outdoor pots.

### 2 • One SKU, Three Sizes

All stickers share identical material, finish, NFC chip and durability. Only the diameter differs:

| Size    | Use case                                                            |
|---------|---------------------------------------------------------------------|
| ø 25 mm | Small pots, herbs, propagation cups, seedling trays                 |
| ø 30 mm | Standard houseplants — default size, most common                    |
| ø 40 mm | Garden plants, outdoor planters, large terracotta pots, raised beds |

The end-customer pack contains all three sizes in a fixed assortment (8 × 25 + 8 × 30 + 4 × 40 = 20 stickers per pack). One pack covers everything from a windowsill basil to a balcony lemon tree.

### 3 • Two-URL Programming — QR vs. NFC

Each sticker carries TWO unique URLs that point to different endpoints in our web app. This is essential for our user experience and is a hard product requirement on the manufacturer.

| Channel               | URL pattern → Behaviour                                                                                                                                                                    |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| QR Code (printed)     | /p/{uuid} → public landing page; user registers / logs in, then sees the plant profile                                                                                                     |
| NFC Chip (programmed) | /nfc/{signed_token} → secure auto-login: if the user is already logged in on the device, they jump straight into the plant profile; otherwise they receive a passwordless magic-link login |

The manufacturer receives a CSV from us with three columns per sticker (uuid, qr\_url, nfc\_url). The QR press uses the qr\_url column for variable printing; the NFC programmer uses the nfc\_url column for chip encoding.

Backend security: the NFC token is signed and bound to a single sticker; it cannot be replayed by anyone who only takes a photo of the QR code. NFC chips must be locked against re-writing after programming.

### 4 • Weather Resistance — Hard Requirement

Because Planum covers outdoor and garden use from day one, every sticker (regardless of size) MUST be fully weatherproof for at least 3 years.

- Permanent water resistance — repeated watering, rain, condensation, full submersion ≥ 24 h
- UV stability — Blue Wool Scale > 6, no fading after 12 months direct sunlight
- Temperature range — -20 °C to +80 °C without adhesive failure or brittleness
- Soil & dirt — wipeable, no permanent staining
- Adhesion on terracotta, glazed ceramic, plastic, glass, powder-coated metal

**Material preference:** PET face material with matte lamination + UV protective overlamine. PVC acceptable as fallback if PET is not in the manufacturer’s portfolio. We are open to a counter-proposal from the supplier provided the durability targets are met.

### 5 • Technical Specifications

| Parameter     | Specification                                             |
|---------------|-----------------------------------------------------------|
| Shape         | Round, die-cut                                            |
| Diameters     | ø 25 mm / ø 30 mm / ø 40 mm — same SKU                    |
| Face material | PET ~80–100 µm (preferred), PVC alternative               |
| Adhesive      | Permanent acrylic, solvent-free, > 8 N/25 mm initial tack |
| Temperature   | -20 °C to +80 °C operational                              |
| Printing      | CMYK 4-colour, ≥ 600 dpi                                  |
| Finishing     | Matte lamination + UV protective overlamine               |

|                                 |                                                                        |
|---------------------------------|------------------------------------------------------------------------|
| <b>Water resistance</b>         | ≥ 24 h immersion + permanent splash                                    |
| <b>Light fastness</b>           | Blue Wool Scale > 6                                                    |
| <b>Durability</b>               | Min. 3 years indoor + outdoor                                          |
| <b>QR code</b>                  | QR Model 2, ECC Level H (30 %), module ≥ 0.5 mm                        |
| <b>QR variable data</b>         | Unique qr_url per sticker (CSV from Planum)                            |
| <b>NFC chip</b>                 | NXP NTAG 213 (180 byte) standard; NTAG 215 (504 byte) optional upgrade |
| <b>NFC frequency / protocol</b> | 13.56 MHz, ISO/IEC 14443A                                              |
| <b>NFC programming</b>          | Pre-programmed unique nfc_url per chip, locked against re-writing      |
| <b>Certifications</b>           | REACH, RoHS, EU 10/2011, NFC chip CE / FCC                             |

## 6 • Packaging — End-Customer Pack

Each end-customer pack contains 20 stickers in a fixed mixed-size assortment, plus printed materials:

- 8 × ø 25 mm (small pots, herbs, propagation)
- 8 × ø 30 mm (standard houseplants)
- 4 × ø 40 mm (garden / outdoor)
- Carrier sheet (A6 or folded A5), pre-arranged by size
- Branded paper sleeve / folded carton with Planum artwork (FSC paper preferred)
- 1 × instruction insert (DIN A6, 120 gsm, double-sided CMYK)
- Plastic-free outer wrap preferred

Shipping carton: 50 end-customer packs per outer carton, with lot number and production date printed on the outside.

Open packaging questions for the manufacturer (asked explicitly in the quotation request): can custom packaging be produced from artwork supplied by Planum, which file formats are accepted (we can deliver AI / PDF/X-1a / EPS / SVG), is there a library of ready-to-customise packaging templates we could choose from, and what colour profile (e.g. Fogra39) and bleed are required.

## 7 • Quantities & Pricing Targets

### 7.1 Pilot Order

| Item                            | Quantity                                   |
|---------------------------------|--------------------------------------------|
| <b>Total stickers (one SKU)</b> | 1,000 pcs                                  |
| <b>End-customer packs</b>       | 50 packs of 20 stickers                    |
| <b>Size mix per pack</b>        | 8 × ø25 mm + 8 × ø30 mm + 4 × ø40 mm       |
| <b>Size mix total</b>           | 400 × ø25 mm + 400 × ø30 mm + 200 × ø40 mm |

|                                     |                                                          |
|-------------------------------------|----------------------------------------------------------|
| <b>Retail price per pack</b>        | €29.00 (hardware) + €10.00/month optional Pro app add-on |
| <b>Target landed cost / sticker</b> | ≤ €0.55 incl. packaging + shipping + duties              |

## 7.2 Follow-Up Orders (planned)

- 5,000 pcs — first re-order after market validation
- 10,000 pcs — Q3 scale-up
- 25,000 pcs — Q4 / holiday season
- 50,000 pcs — annual run-rate target

## 8 • Manufacturer Requirements

### 8.1 Mandatory

- Acceptance of CSV with 1,000+ unique entries (uuid, qr\_url, nfc\_url) for variable QR printing AND NFC programming
- NFC chips pre-programmed with unique nfc\_url per UUID, locked against re-writing
- Fully assembled test pack (1–2 end-customer sets) before main production
- Digital proof / print preview before sign-off
- Test reports: water immersion, UV exposure, peel adhesion on terracotta
- Custom packaging from Planum artwork — sleeve, carrier, instruction insert
- Shipping options: DHL/FedEx/UPS air courier (DDP Germany) AND CIF Hamburg quoted separately

### 8.2 Preferred

- FSC-certified paper for sleeves and carrier; plastic-free outer wrap
- In-house warehousing / fulfilment option for follow-up orders
- Documented experience with combined QR + NFC stickers (reference projects)
- NDA on request

## 9 • Brand Colours

|                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                      |                                                                                       |                                                                                       |
|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
|  |  |  |  |  |  |  |
| <b>Forest</b><br>#1F4D2C                                                            | <b>Leaf</b><br>#2D5F3F                                                              | <b>Sage</b><br>#87A878                                                              | <b>Mist</b><br>#C7D7C2                                                              | <b>Cream</b><br>#F5F1E8                                                              | <b>Paper</b><br>#FBFAF6                                                               | <b>Ink</b><br>#15241B                                                                 |

Final print-ready vector files (AI / PDF/X-1a) will be delivered after the manufacturer has confirmed accepted file formats and required colour profile.

## 10 • Shipping & Payment Terms

|                                  |                                                              |
|----------------------------------|--------------------------------------------------------------|
| <b>Sample / test pack</b>        | Within 14 days of order confirmation, via DHL Express        |
| <b>Main production lead time</b> | Max. 30 days from sample approval                            |
| <b>Pilot shipping (~2 kg)</b>    | Air courier preferred (DDP Germany incl. customs + 19 % VAT) |

|                           |                                                             |
|---------------------------|-------------------------------------------------------------|
| <b>Follow-up shipping</b> | Sea LCL CIF Hamburg + air courier alternatives, both quoted |
| <b>Payment</b>            | Alibaba Trade Assurance preferred; alternative 30 % / 70 %  |
| <b>Warranty</b>           | Replacement or credit for defect rates > 2 %                |